Karlshochschule
International University

Swapfiets

# SUMMARY REPORT

Swapfiets Team 1

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# ABOUT SWAPFIETS

## What is Swapfiets?

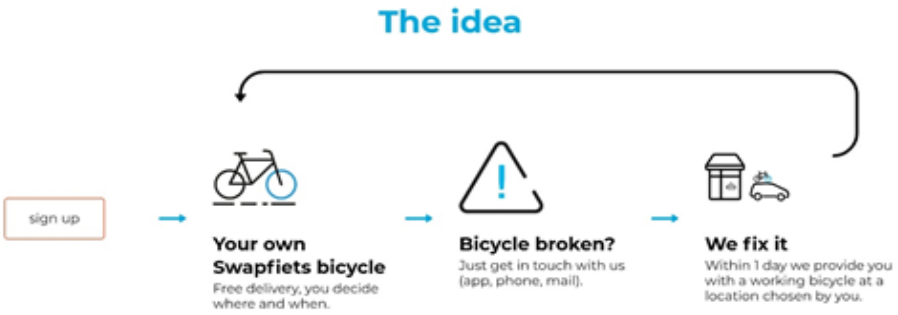
In 2014, Delft, the Netherlands, three technical students founded and launched the idea of Swapfiets. **Swapfiets** is a bicycle subscription, not a bicycle rental company. This Dutch company offers a robust bike as a service, in which they lease to customers for medium to long-term periods. These bicycles with their distinctive blue front tire can be seen zooming around in more than 50 cities located in the Netherlands, Belgium, Denmark, and Germany.

## The Concept

This bicycle subscription concept couldn't be simpler; you can purchase your subscription online or in-store. You sign up for the city which you are living in, choose a bike with the following availability choices, provide a few details, such as your height to determine bike size and seat height, and then either pick up your Swapfiets bike at the local city shop or have your bicycle delivered to your home in no time! With this 'swapping' concept, for when your bicycle breaks down and needs to be urgently exchanged ('swapped'), Swapfiets is there for you with a working one.

An always working bicycle, with no hassle.

Swapfiets continue their massive growth by further introducing their 'Swapfiets Business' concept, in which they provide Swapfiets bicycle packages for companies. From company pool bikes, for hotels and their guests, and personal subscriptions for employees, Swapfiets seeks to suit every company[1].




Source: Swapfiets Website

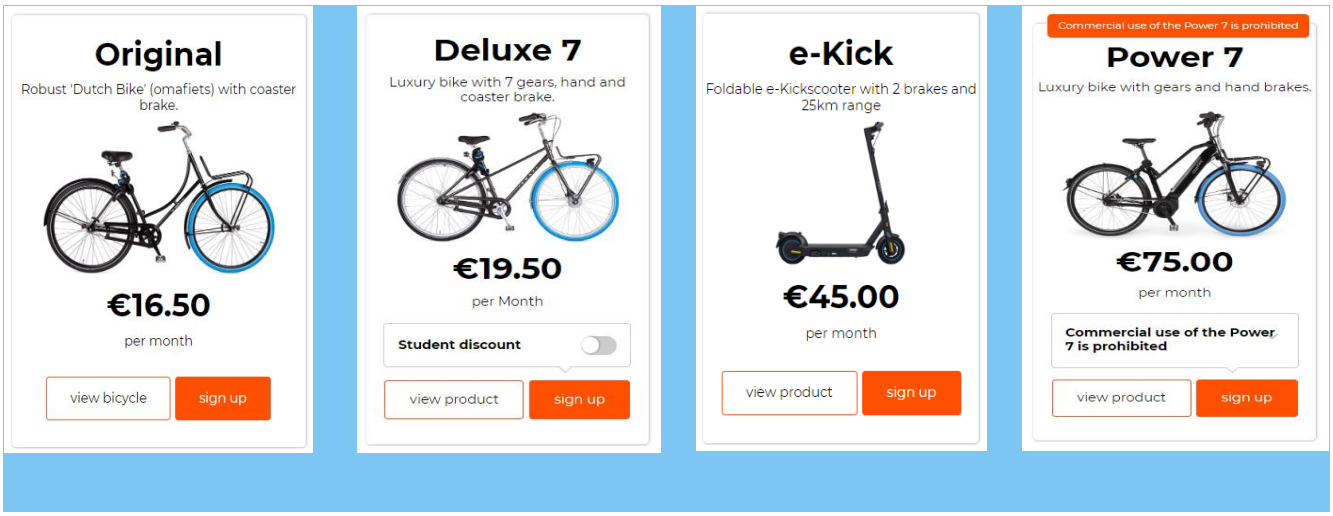
[1] https://business.swapfiets.com/en-DE

## The Promise

Swapfiets bike subscription always works for your everyday commute. You pay €19,50 (€17,50 euros with a student discount) per month. Swapfiets fixes your bike when in need of repairs, and if your bike is stolen and locked properly, you pay a €60,00 deductible, and the remaining costs are covered by Swapfiets cycle insurance. The subscription is flexible, and the termination period is at least one month.[2]

## The Models

There are three bike models: original, deluxe 7, and electric. Each bike is provided with their own lock and own keys[3]. Availability of the bikes are depended on each city. Below are the Swapfiets offered.




Source: Swapfiets websites (The Netherlands [NL], Germany [DE], Denmark [DK], Belgium [BE]).

[2] https://swapfiets.de/en/

[3] https://johnnyafrica.com/swapfiets-bike-review/

## World's Largest Subscription Bike Service

Since its inception in 2014, Swapfiets has grown at least 500% every year. By the end of 2017, the had 20,000 customers, and then by the end of 2019, they had nearly 100,000 customers[4]. Their international growth is now seeking new markets in Australia, Japan and the United States, in Oregon and Colorado[5].

While Swapfiets becomes a global player, they stick with the 'local for local' principle, in which enables them for faster and quality results when entering new cities and countries. The unique company has further developed and advanced against competitors due to its concept of 'bike as a service', and the Netflix-style subscription enables customers to recognize the simplicity of blue-tire phenomenon.

Swapfiets gifts their 100,000th customer with a golden bike. (Silicon Canals, 2019)

[4] https://siliconcanals.com/news/3-unique-ways-dutch-bike-startup-swapfiets-is-winning-customers-and-growing-business-rapidly/

[5] https://www.bloomberg.com/news/articles/2019-08-22/swapfiets-bike-subscription-service-prepares-for-global-expansion

# OUR IPRO PROJECT

## Why Karlsruhe?

*"Adding one option to the mobility choice of people in Karlsruhe"*

Since January of 2019, Swapfiets opened a location in Karlsruhe, Germany. The Dutch company is located perfectly in the city centre.

Karlsruhe was recognized as a bike-friendly municipality in 2017[6], therefore, eventually, the appealing city would attract Swapfiets aspiration to be represented in this lively city.

## Our Task

As the trend of bike sharing, swapping, and renting intensifies, our team's project task is to create a new marketing strategy, as our goal is to increase the customer-base of Swapfiets in Karlsruhe, especially during the winter months.

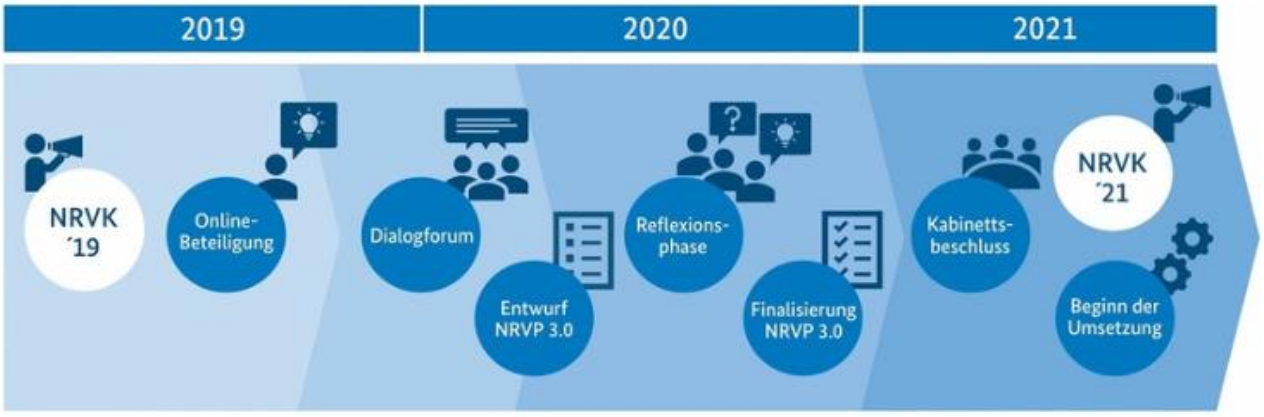
[6] https://www.ka-news.de/region/karlsruhe/fahrradfahren-karlsruhe./Sharing-kommt-auch-bei-Fahrraedern-gut-an-In-Karlsruhe-finden-sich-immer-mehr-Leih-Angebote;art6066,2329575

# CYCLING IN GERMANY

Germany is amongst the countries that have developed a national strategy to promote cycling, as cycling accounts for a rising share of transportation volume. According to the German Federal Ministry of Transport and Digital Infrastructure, more than 80% of the Germans use bicycles, becoming one of the leading cycling nations. With the current, future transport and social challenges, cycling promotes positive impact on the environment, quality of life in towns and cities, and people's health[7].

## Federal Support for Cycling Mobility

Since 2002, Germany has implemented a National Cycling Plan, which is the Federal Government's strategy for the promotion of cycling. As BMVI is currently in the process of developing the National Cycling Plan 3.0, it is going to be applied in 2021. Below are the further developmental plans of NCP:[8]




Source: BMVI

The National Cycling Plan 3.0 identifies and focuses on the diverse objectives for the future cycling promotion in Germany, such as safety, security and digitalization. The current funding of the BMVI is an annual 25 million euros for cycle superhighways, in which federal states and local authorities are responsible of[9].

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[7]National Cycling Plan 2020: Joining forces to evolve cycling. October 2012. Federal Ministry of Transport, Building, and Urban Development. [Translation]. 2nd Edition. Division UI 31 – Passenger Transport, Public Transport Systems, Cycling.

[8]https://www.bmvi.de/SharedDocs/EN/Dossier/Cycling/cycling-in-germany-is-booming.html

[9]https://www.bmvi.de/SharedDocs/EN/Dossier/Cycling/cycling-in-germany-is-booming.html

# Sharing economy

In Germany, bicycles are not only bought but also increasingly shared. This corresponds to the current trend of the "Sharing economy". This popular concept is concerned with

- Owning less stuff
- Sharing, renting or leasing products rather than individual product ownership
- Collaborative consumption, thus it is a minimalist movement.[10]

This comes along with sustainable behaviour.[11]Participants of this consumer trend are driven by the feeling that they have some control over their relationships with companies. In the psychological world, it is called “Self-actualization", which is at the top of Maslow’s hierarchy of needs.[12]

[10]https://www.forbes.com/sites/bernardmarr/2016/10/21/the-sharing-economy-what-it-is-examples-and-how-big-data-platforms-and-algorithms-fuel/
[11]https://ec.europa.eu/transport/themes/urban/cycling/guidance-cycling-projects-eu/cycling-measure/bicycle-sharing_en
[12]https://www.skyword.com/contentstandard/the-consumer-psychology-behind-the-sharing-economy-what-marketers-must-know/

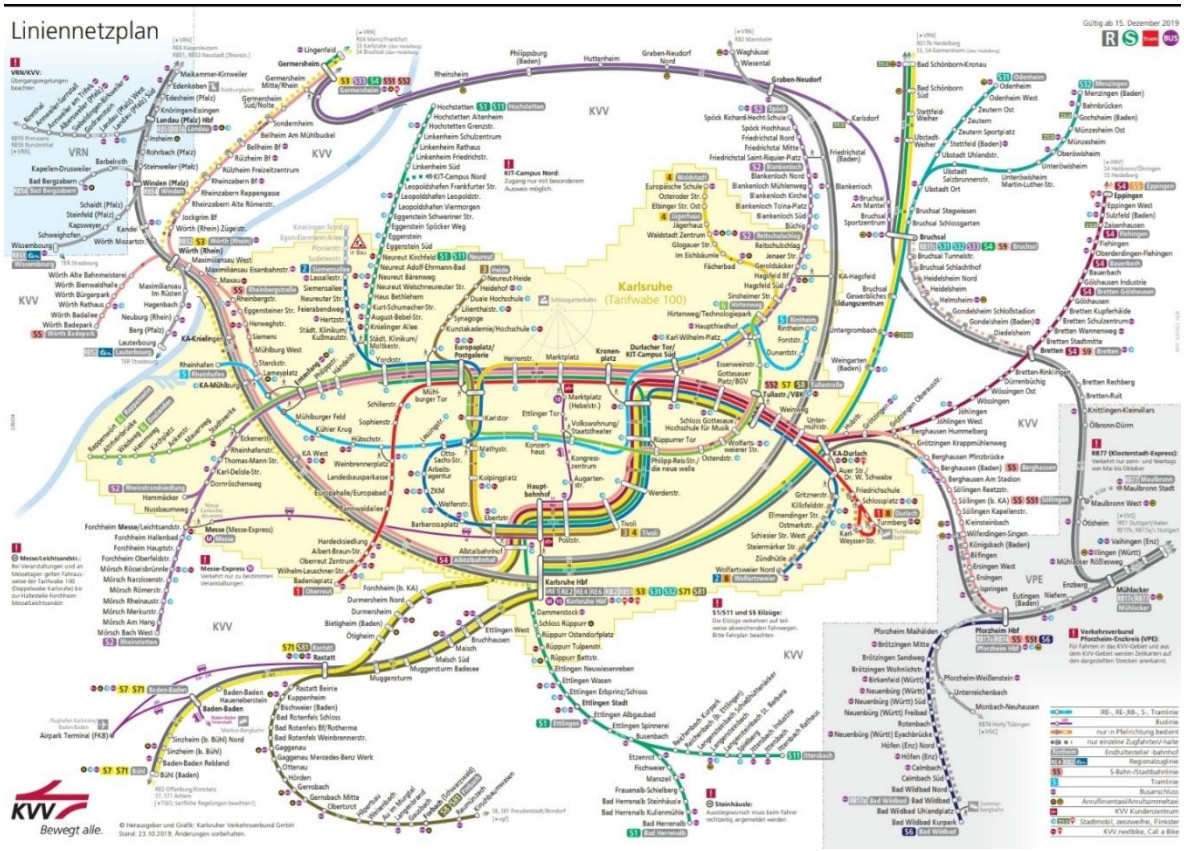
# Mobility market Karlsruhe

Having a closer look at the mobility market and the usage of different transportation means in Karlsruhe, there are several outstanding findings that have to be stressed in terms of Swapfiets‘ indirect and direct competitors.

## Indirect competitors

1) ***Buses &trams& light + regional railsof KVV (Karlsruher Verkehrsverbund)***
    - The KVV = third largest transport association in Baden-Württemberg[13]
    - Since 2014:morethan 170 million people a year have been using the KVV service[14]
    - Unified transport&tariffsystemfor262lineswithinthe KVV
    - Types & number of lines
        - 12 regional railway lines
        - 15 light rail lines (Stadtbahnlinien)
        - 7 tramlines  (Straßenbahnlinien)

[13] https://www.kvv.de/unternehmen.html
[14] https://www.kvv.de/unternehmen/zahlen-fakten.html




- 228 bus lines (Buslinien)[15]

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- Special train / tram tickets for bikes are necessary

➔ From Monday to Friday between 6 a.m. and 9 a.m., a bicycle can be taken on the trains of Deutsche Bahn AG, Abellio Rail Baden-Württemberg GmbH and Go-Ahead Baden-Württemberg GmbH, the AVG light rail vehicles and the VBK trams, if a bicycle card (price for KVV network: €2,60)has been purchased. Each passenger can only take one bike with him / her and the bicycle card is only valid in conjunction with a valid KVV ticket. [17]

[15] https://www.kvv.de/unternehmen/zahlen-fakten/strukturdaten.html
[16] https://www.kvv.de/liniennetz/schiene.html
[17] https://www.kvv.de/fahrkarten/fahrkarten-preise/radfahrer.html

## 2) ***Motorized individual traffic***

- Concerning the main means of transport in Karlsruhe,it can be noticed that 34,4% of the population there mainly uses motorized individual transportation means (including car and motorcycle)
- Therefore, this transportation mean is the most popular one compared to public transportation, bikes and walking

**Tab. 28 Hauptverkehrsmittel nach Alter, Geschlecht und Mobilitätseinschränkung**

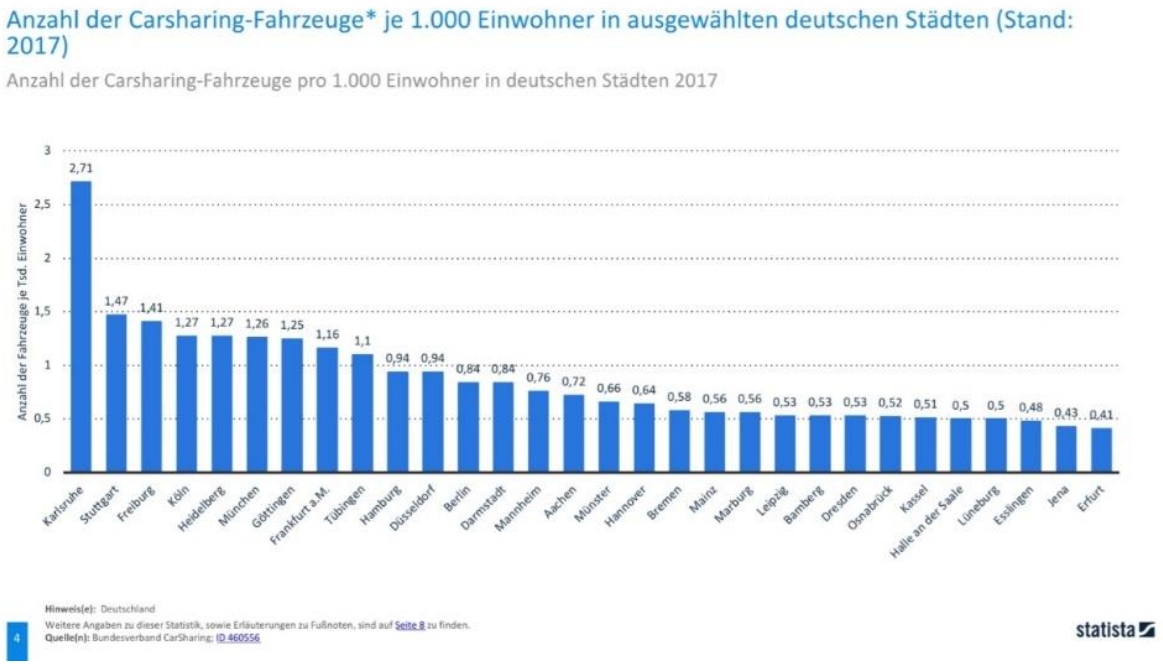
n = ungewichtete Fallzahl; ohne sonstige Verkehrsmittel; Definition Hauptverkehrsmittel siehe Tab. 2; nur Karlsruhe

| | gesamt Karlsruhe | Geschlecht | | Alter | | | | Mobilitätseinschränkung | |
|---|---|---|---|---|---|---|---|---|---|
| | | männlich | weiblich | bis 16 | 17 - 34 | 35 - 59 | 60+ | nein | ja |
| zu Fuß | 24,4% | 21,7% | 27,1% | 29,3% | 20,7% | 20,6% | 32,5% | 24,1% | 28,0% |
| Fahrrad | 24,5% | 27,9% | 21,2% | 27,5% | 31,5% | 24,7% | 14,8% | 25,6% | 10,4% |
| MIV als Mitfahrer/-in | 8,7% | 6,8% | 10,6% | 24,6% | 8,9% | 6,3% | 5,3% | 8,6% | 9,7% |
| MIV als Fahrer/-in | 25,7% | 30,2% | 21,3% | 0,0% | 17,8% | 36,2% | 29,7% | 25,8% | 24,9% |
| ÖV-Nahverkehr | 16,1% | 12,5% | 19,5% | 18,4% | 20,1% | 11,4% | 17,6% | 15,2% | 27,0% |
| ÖV-Fernverkehr | 0,6% | 0,9% | 0,3% | 0,1% | 0,9% | 0,8% | 0,0% | 0,6% | 0,0% |
| n (Wege) | 10843 | 5254 | 5589 | 1955 | 1971 | 4802 | 2115 | 10261 | 582 |

[18]

- Car-sharing = very popular and an increasing trend in Karlsruhe
  - ➔ According to the Bundesverband Carsharing, there are about 2.71 car-sharing vehicles per 1,000 inhabitants in Karlsruhe (1st place in Germany, 2017). This puts Karlsruhe in first place nationwide and hence, it can be called the "carsharing capital".

[18] https://www.karlsruhe.de/b3/verkehr.de



[19]

This is mainly thanks to the car-sharing company **Stadtmobil**, which has a very well developed network of stations in Karlsruhe. Stadtmobil has a total of 240 locations in Karlsruhe. Deutsche Bahn is also getting involved with its own carsharing service called **Flinkster**. However, Flinkster canonlybefound at Karlsruhe mainstation.[20]

### *3) <u>E-scooter</u>*

So far, there are two providers in Karlsruhe who rent out e-scooters:

I. **Voi**

- 20.09.2019: Swedish company VOI = first rental company for e-scooters in Karlsruhe
- Costs: one euro entry fee plus 15 cents per minute per ride[21]

II. **Tier**

- Since 12.12.2019: e-scooters of Tier have been available in Karlsruhe
- To use scooters: download Tier app, start scooter by scanning QR code on handlebars

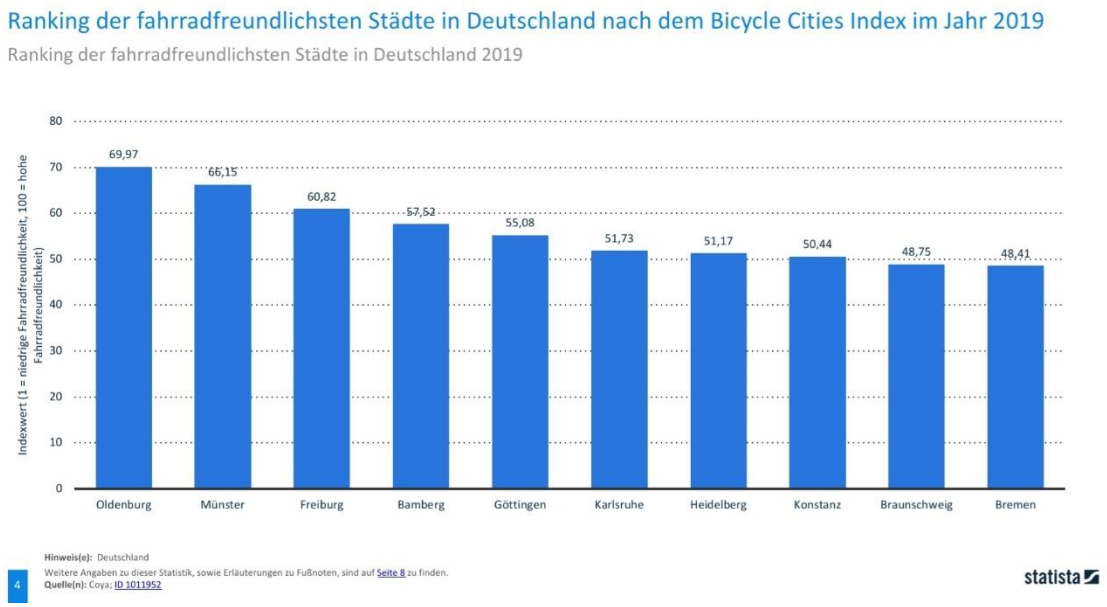
[19] https://de.statista.com/statistik/daten/studie/460556/umfrage/carsharing-anzahl-der-fahrzeuge-in-deutschen-staedten-pro-tausend-einwohner/
[20] https://www.carsharing-news.de/carsharing-karlsruhe/
[21] https://bnn.de/nachrichten/verkehr/erster-verleiher-fuer-e-scooter-geht-in-karlsruhe-an-den-start

- After usage: scooter can be parked flexibly at suitable places
- Costs: activation fee of one euro per ride plus 15 cents per minute[22]

## Bike market

Karlsruhe is an extremely bicycle-friendly city, which can be seen in a rankingoftheinsurancecompanyCoyawhoevaluatedtheoverallcyclingconditionsof German cities. The following main categories were examined: The weather, bicycle use, crime and safety, bicycle sharing and events. The resulting Bicycle Cities Index ranges from 1 to 100, the higher the value, the more bicycle-friendly the city.

According to this statistic, Karlsruhe ranks sixth.



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Moreover, Karlsruhe received the state award as "bicycle-friendly city" in 2011 and again in 2017.[24] This is mainly possible thanks to the policy of the city of Karlsruhe regarding bicycle traffic, where by the following points actively contribute to bicycle-friendliness:

- 20-point program to promote biking
- Cycling network (city + district routes) consisting of for example special bike streets
- Cycling forum where goals of cycling policies & plans are discussed

[22] https://bnn.de/lokales/karlsruhe/zweiter-e-scooter-anbieter-in-karlsruhe-gestartet
[23] https://de.statista.com/statistik/daten/studie/1011952/umfrage/ranking-der-fahrradfreundlichsten-staedte-deutschlands/
[24] https://www.karlsruhe.de/b3/verkehr/radverkehr/radzaehler.de

- Bicycle station at main station→Totally: over1.000 bicycles can be parked there
- Winter road maintenance to reliably ensure a continuous, ridable network[25]

**Bike shops**

In Karlsruhe, there are several bike shops where people can buy almost every type of bicycles like mountain bike, racing bike, cargo bike, recumbent bike, children's bike, all terrain bike, trekking bike. →If people are interested in using one of these, they won't rent if from Swapfiets since they only offer city bikes / Holland bikes& e-bikes

## Direct competitors

1) ***KVV NextBike[26]***
    - The bike sharing system KVV nextbike is another alternative to bus, train and car
    - For usage: register via app, service hotline or website, borrow at any station by scanning he QR code
    - After use: Return the bike at any KVV next bike station
    - Cost of basic bike: First 30 minutes €1, further 30 minutes each €1, maximum daily rate €9
    - Costs of E-bike: double the basic rate

2) ***DB Call a bike***
    - For usage: register & borrow bike via app, customer card or terminal
    - After use: Return the bike at a station[27]
    - Standard rate: 30 minutes each, daily rate €15
    - Discount rate: 30 minutes each, daily rate €12 (e.g. for students)[28]

[25] https://www.karlsruhe.de/b3/verkehr/radverkehr.de
[26] https://www.kvv.de/service/angebote-aktionen/kvvnextbike.html
[27] https://www.callabike-interaktiv.de/de
[28] https://www.callabike-interaktiv.de/de/tarife

# Market Segmentation

## *1. Market segmentation based on psychographics*

As a first step in market segmentation, we considered the characteristics of potential customers for the product. We found that the Swapfiets bike is a gender neutral product, taking into account that there is no significant **difference regarding cycling rate between men and women in Karlsruhe:**

**Tab. 28 Hauptverkehrsmittel nach Alter, Geschlecht und Mobilitätseinschränkung**

n = ungewichtete Fallzahl; ohne sonstige Verkehrsmittel; Definition Hauptverkehrsmittel siehe Tab. 2; nur Karlsruhe

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| n (Wege) | 10843 | 5254 | 5589 | 1955 | 1971 | 4802 | 2115 | 10261 | 582 |

[29]

Furthermore, it can be noted that there is **no significant difference regarding cycling rate between the different age groups of Karlsruhe:**

**Tab. 28 Hauptverkehrsmittel nach Alter, Geschlecht und Mobilitätseinschränkung**

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[30]

However, there is one exception looking at the product: Children are not able to use the bike because their height.

Moreover, we found that it makes sense to **subdivide the bike market into distinct segments that possess similar wants and needs** regarding cycling and more specifically about bike renting. Thereby, the overall aim was to create a marketing mix that accurately matches the expectations of Swapfiets customers in the targeted bike market segments. Thus, out of the four pillars of market segmentation

[29] https://www.karlsruhe.de/b3/verkehr/HF_sections/rightColumn/ZZo2Feyo0MKXeI/ZZo2AIoFgpznG4/Ergebnisbericht_Karlsruhe_Endversion_5cac6d1bdcd25.pdf

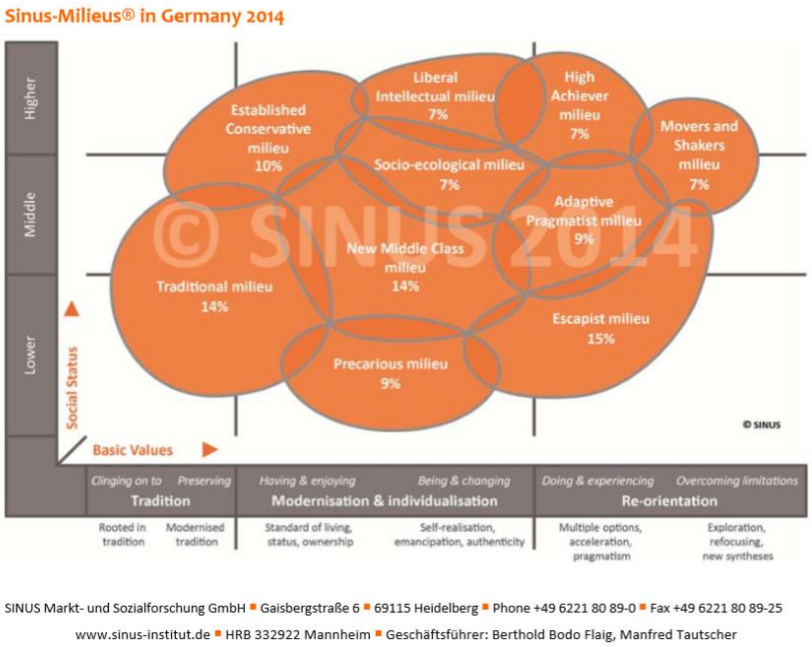
[30] https://www.karlsruhe.de/b3/verkehr/HF_sections/rightColumn/ZZo2Feyo0MKXeI/ZZo2AIoFgpznG4/Ergebnisbericht_Karlsruhe Endversion_5cac6d1bdcd25.pdf

(Geographic, Demographic, Psychographic, Behavioral)[31], we focused on the **psychographic dimension** of it.

## *2. Applying Sinus-milieus model for segmentation*

Based on that, we decided to choose the **Sinus-milieus model** because it "offers a true-to-life image of the socio-cultural diversity of the modern world."[32] It describes **people´s values, lifestyles, attitudes and orientations.** With the help of the model, we were able to explore the life worlds of Swapfiets potential target groups.

## *3. Analysing milieus regarding their general characteristics[33]*




### *1. Established Conservative milieu*

- Responsibility& success ethic
- Aspirations of exclusivity

### *2. Liberal Intellectual milieu*

- Fundamentally liberal &enlightened educational elite
- Array of intellectual interests

[31] https://www.emailmonday.com/smart-email-marketing-segmentation-the-art-of/
[32] https://www.sinus-institut.de/en/sinus-solutions/sinus-milieus/
[33] https://www.researchgate.net/figure/Sinus-Milieus-3-R-in-Germany-2011-source-SINUS-Institut-Heidelberg-Figure-3-Sinus_fig3_258685920

*3. High Achiever milieu*

- Self-image as avant-garde when it comes to consumption

*4. Movers and Shakers milieu*

- Culturally & geographically mobile
- Always looking for new challenges & change
- Search of new boundaries

*5. New Middle Class milieu*

- Seeking to lead a secure & harmonious existence

*6. Adaptive Pragmatist milieu*

- Motivated & prepared to adapt
- Pragmatic outlook on life

*7. Socio-ecological*

- Pronounced ecological & social conscience
- Globalisation sceptics

*8. Traditional milieu*

- Tradition & order-loving old generation

*9. Precarious milieu*

- Lower class in search of orientation
- Withdrawal into own social environment

*10. Escapist milieu*

- Fun, adventure- &experience-oriented
- Modern
- Living in the here & now
- Carefree

***4. General research →people´s wants & needs (motivators) concerning bike renting & cycling in general →connecting them to milieus***

| **Wants&needs** | **Milieus** |
|---|---|
| ➢ Seekforsafety&security | New middleclassmilieu |
| ➢ Less responsibility (not take care of bike oneself) &fun &experience | Escapistmilieu |
| ➢ Variety of mobility forms (flexibility) | Moversand Shakers milieu |
| ➢ Lesspossession (minimalism) →sharingeconomy | High Achievermilieu |
| ➢ Cheaper price (bike = cheapest alternative) | Precariousmilieu |
| ➢ Sustainability & healthy lifestyle (stay fit) | Socio-ecologicalmilieu |
| *(no match between motivator and milieu)* | Liberal intellectualmilieu |
| *(no match between motivator and milieu)* | Adaptive pragmatistmilieu |
| *(no match between motivator and milieu)* | Establishedconservativemilieu |
| *(no match between motivator and milieu)* | Traditional milieu |

***5. Analysing components of Swapfiets' bike rental service →assigning them to milieus***

| **Components ofSwapfiets'service**[34] | **Milieus** |
|---|---|
| ➢ Alwaysworking bike<br>➢ Fixed monthlyfee | New middleclassmilieu |
| ➢ Nohasslewithrepair | Escapistmilieu |
| ➢ Monthlysubscription = flexibility | Moversand Shakers milieu |
| ➢ Rentinginsteadofowning | High Achievermilieu |
| ➢ Discounts | Precariousmilieu |
| *(no match between component and milieu)* | Socio-ecologicalmilieu |

34https://swapfiets.de/

## ***6. Result: Detailed description of milieus (milieus' characteristics & motivators & components of Swapfiets bike rental service)***

For the distinct milieus, there is a different relevance of Swapfiets' bike rental service components (criteria).The most relevant criteria were assigned to the milieus according to their general characteristics. Moreover, people have various wants and needs regarding bike renting and cycling in general, which were assigned to the groups of the Sinus milieu model.

*New middle class milieu*

The new middle class milieu is characterized by seeking for security and harmonious existence. This goes along with the general need for safety when it comes to cycling, where bythe ADFC stresses that many cyclists feel increasingly insecure.[35]Thus, Swapfiets can offer with its service an always working bike in perfect condition (brakes, lightening), which results in more safety while riding a bike. Moreover, the fixed monthly fee by Swapfiets results in clearly calculable costs, including all repair costs, insurance and double lock from Axa (=protection against bike theft). This means safety in terms of being sure that there are no unexpected costs for the rented bike.

*Movers and Shakers milieu*

This group of people is always looking for new challenges and change. They are culturally and geographically mobile, which leads to a need for flexibility regarding mobility forms they can use. This need for flexibility is fulfilled by Swapfiets since customers can rent the bike on a monthly basis, which enhances their flexibility.

*High Achiever milieu*

The Higher Achiever milieu has a self-image as avant-garde (=pioneers, pacemakers, being progressive[36]) when it comes to consumption. In the society of today, this implies that one is in favour of less consumerism and the minimalist movement. Thus, these people strive for possessing fewer things and rather sharing them This means that they are participants of the sharing economy, which focuses on collaborative and more responsible consumption where social sharing of goods is

[35]https://www.tagesspiegel.de/wirtschaft/verkaufszahlen-warum-kommt-die-verkehrswende-bei-der-fahrradbranche-nicht-an/24084332.html
[36]https://www.duden.de/rechtschreibung/Avantgarde

common[37]. Swapfiets' service exactly fits in the sharing economy trend with its concept of renting bikes, rather than owning them.

*Precarious milieu*

This group of people belongs to the lower class, which is in search of orientation. Their main need is a cheap mean of transportation that is given by choosing a bike due to the low cost of maintenance compared to other modes of transport. Swapfiets serves this need by charging a fixed monthly fee, whereby the costs are clearly calculable compared to owning a bike. Furthermore, Swapfiets offers several discounts (e.g. students, welcome package and sweepstakes online), which makes the service attractive for this milieu.

*Escapist milieu*

The people belonging to this milieu are carefree, fun- and experience-oriented. Therefore, they do not want to worry about bike maintenance, which means being less exhausted and more comfortable. With Swapfiets' offer, the customer has no hassle with repairs and thereby saves time. Through this convenience and recklessness, the milieu can focus on fun, experience and pleasure. Since this milieu is modern low-middle class, the price aspect of a bike is also crucial for them.

*Socio-ecological milieu*

The socio-ecological milieu is characterized by pronounced ecological conscience and globalisation sceptics. Hence, the motivator of these people concerning cycling is to be environmentally friendly. By leading a sustainable lifestyle, the environment is affected positively (less pollution) and thereby people live healthier.[38]

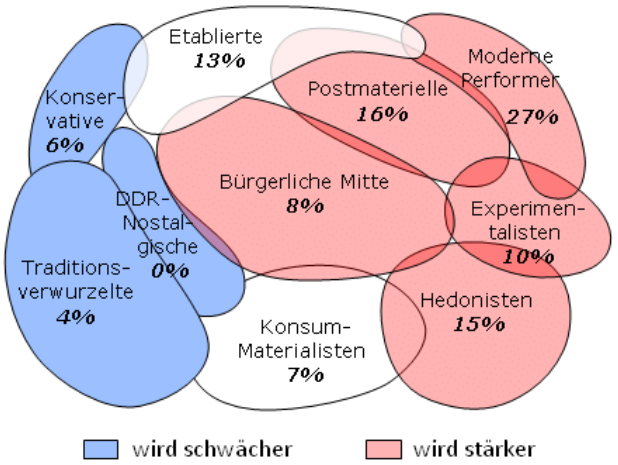
## *7. Ranking of milieus: Highest market potential= ranked highest→target groups*

In order to evaluate **the market potential** of the six above mentioned milieus, **two criteria** were taken into account:

- The **degree of matching / fit** between milieus and Swapfiets components (more matches = higher market potential)
- **Share of the milieus in the population** of Karlsruhe (in percentage)

[37] https://ec.europa.eu/transport/themes/urban/cycling/guidance-cycling-projects-eu/cycling-measure/bicycle-sharing_en
[38] https://www.unthsc.edu/operations/sustainability/sustainability-and-healthcare/

*Sinus Milieus in Karlsruhe in 2020 (please note that milieus names are slightly different):*



[39]

*Ranking*

| **Milieu** | **Match withcomponents** | **Share ofmilieu in Karlsruhe** | **Ranking** |
|---|---|---|---|
| Escapist | 2 matches | 15% | 1 |
| New middleclass | 2 matches | 8% | 2 |
| High Achiever | 1 match | 27% | 3 |
| Moverand Shakers | 1 match | 10% | 4 |
| Precarious | 1 match | 7% | 5 |
| Socio-ecological | Nomatch | 16% | 6 |

The four target groups that will be taken into account are

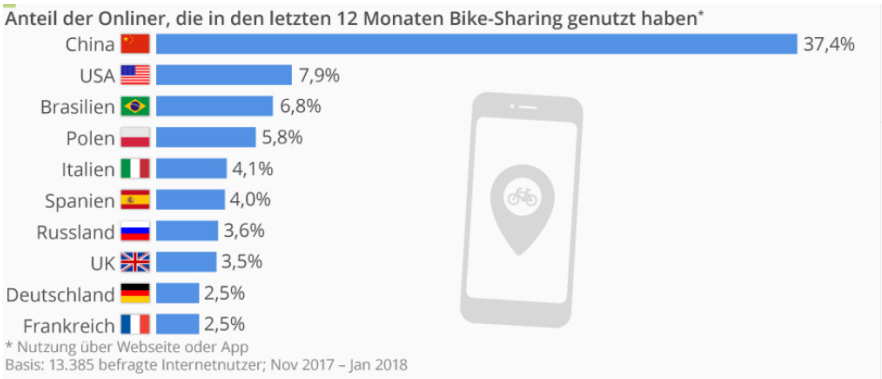
- Escapist milieu
- New middle class milieu
- High Achiever milieu
- Movers and Shakers milieu

[39] https://www.researchgate.net/figure/Abbildung-9-Sinus-Milieus-in-Karlsruhe-im-Jahr-2020_fig2_45828976

The socio-ecological milieu is not considered as a target group as there is no match between the components of Swapfiets' bike rental service. Moreover, we put no special emphasis on the precarious milieu as target group because in comparison to the other milieus, the market potential is not that big in Karlsruhe and there is only one match.

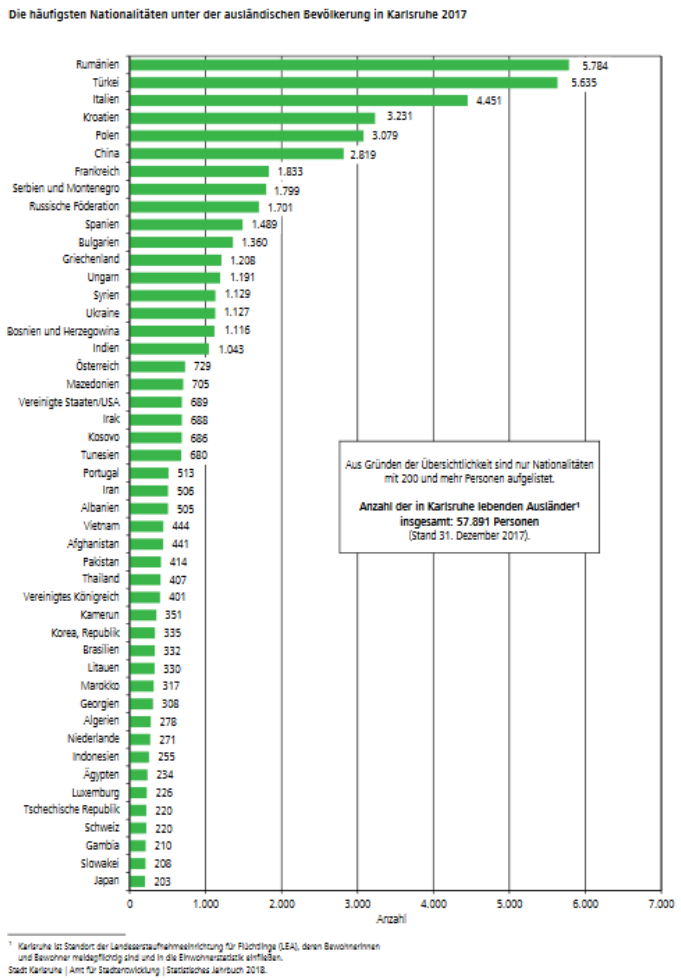
*Target Chinese students living Karlsruhe (High Achiever)*

After defining the four target groups, we took another look at the results of our general research and tried to characterize the four milieus mentioned above a little more specific. When we looked again at the **High Achiever milieu**, we found the following: With regard to the aspect "sharing economy / less possession" (in relation to this group), it can be noticed that China is worldwide by far the leader in terms of bike sharing.




According to this survey, 37.4% of the Chinese population have used bike sharing services in the last 12 months[40], which means that there already is a **bike-sharing culture within the Chinese population**. Through further research it was found out that the **Chinese citizens are one of the most common nationalities in Karlsruhe** among the foreign population (in sixth place).

[40] https://de.statista.com/infografik/15936/bike-sharing-weltweit-und-in-deutschland/

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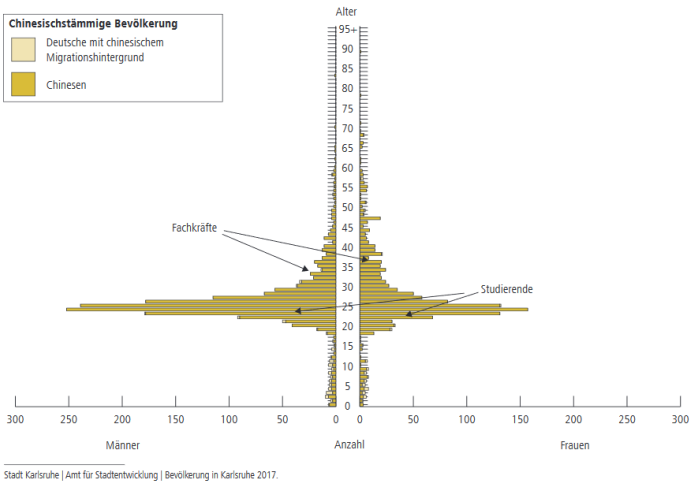
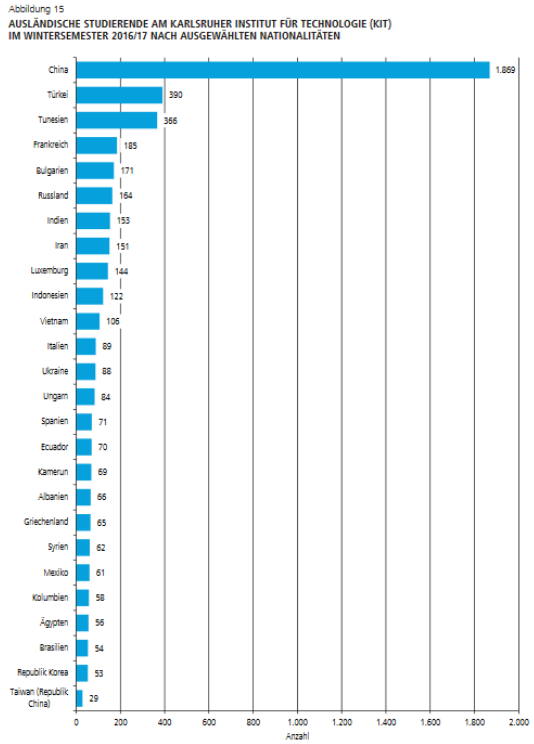
Then, regarding the age distribution it is outstanding that there are **predominantly young Chinese citizens**(between 20-30 years old) and **most of them are students.**




42

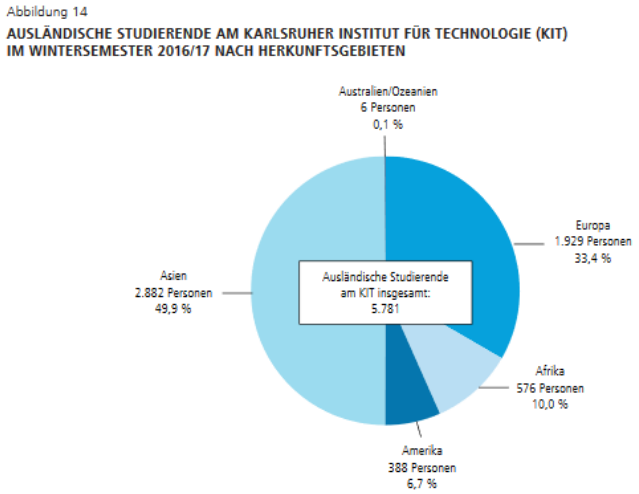
[41]https://web3.karlsruhe.de/Stadtentwicklung/statistik/pdf/2018/2018-jahrbuch.pdf
[42]https://web3.karlsruhe.de/Stadtentwicklung/statistik/pdf/2017/2017-bevoelkerung-jahresbericht.pdf

Most of them are at the “Karlsruher Institut für Technologie” (KIT). There, they make up the largest share of foreign students (total number = 1.869 Chinese students).



[43]

It is outstanding that the number of Chinese students at KIT is similar to the amount of European students (1.929 Europeans), which emphasizes the enormous market potential of targeting Chinese students.



[44]

[43] https://web3.karlsruhe.de/Stadtentwicklung/statistik/pdf/2016/2016_2017-studierende.pdf
[44] https://web3.karlsruhe.de/Stadtentwicklung/statistik/pdf/2016/2016_2017-studierende.pdf

*Target exchange students living in Karlsruhe*

When we looked again at the **Movers and Shakers milieu**, we found the following: With regard to the aspect "culturally and geographically mobile+ flexibility" (in relation to this group), we detected a correlation between this milieu and the **exchange students** living in Karlsruhe. We consider that for this group of people, Swapfiets' concept of renting a bike on a monthly basis is ideal because they are only in Karlsruhe for a few months. Consequently, it is not worth it for them to buy an own bike because first they have to buy it, then they have to invest time and money in possible repairs and then they have to take care of the sale just before they leave. When taking **indirect competitors** into account, it can be noticed that public transport would be the most probable transportation alternative for exchange students besides bikes. However, public transportation in Karlsruhe is comparatively expensive (KVV semester tickets for students currently costs 160.80Euro[45]) to the bike rental service provided by Swapfiets, plus the student discount (exchange students are on average for 5 months in Karlsruhe: 5 * 17.50Euro = 87.50Euro[46]). Moreover, exchange students are not that flexible with public transportation due to dead public transportation zones, especially during night and the weekends.

[45] https://www.kvv.de/fahrkarten/fahrkarten-preise/schueler-studenten/studikarte.html
[46] https://swapfiets.de/offer/karlsruhe

## Quantitative study: Survey

### Aim

The aim of this study is to see if there is a correlation in different motivators of cycling on the Milieu of an individual. Additionally, different possible trends were taken into account to gain an understanding of the population in Karlsruhe, Germany. The possible factors that were taken into account include age, occupation and nationality. Also, to improve winter cycling and gain a competitive advantage in the market, suggestions for the winter season were also taken.

### Study Method

The study method used is a survey. A survey is said to be one of the easier and quicker methods that also creates relative valuable data to represent a certain group or population, in this case Karlsruhe. When individuals in Karlsruhe are the target group, then the study has to be conducted in that location. This makes Karlsruhe part of a control (also known as a constant), meaning that this will not be changed throughout the survey. Also due to the lack of time and preference of quantity rather than quality of data, the survey will be conducted with multiple surveyors. The study was conducted in the city center of Karlsruhe, but the specific location changed multiple times to gain a greater number of respondents.

### Sampling Technique

The sampling technique used for this survey is “simple random sampling”, meaning that participants were not chosen in an orderly manner. In our case they were randomly stopped in the streets of Karlsruhe. This method is commonly used to eliminate large scales of researcher, sampling and selection bias. However, due to its simplicity, many downsides exist such as not being able to look at groups in detail, large generalisation of the population and it’s difficult to calculate the margin of error. While it is one of the most straightforward sampling methods, many disadvantages still arise. This sampling method does not account for such problems and makes it difficult to estimate the errors that exist within the findings. However, in the situation of the Swapfiets group, it was ideal to use this sampling technique to both save time and gain “random” results of individuals in Karlsruhe for the purpose of representing the general public.

**Design**

| Questions | Reasonofchoice |
|---|---|
| Are you interested in a monthly subscription that offers an always working bike including maintenance and repair? | To know how many percentages of participants are interested in the concept of monthly subscription for renting a bike (Swapfiets offer). |
| **Main Motivator** | |
| What would be your main motivator for this bike subscription? | To determine if each motivator aligns with the sinus milieus we assigned. |
| **Winter Cycling** | |
| When cycling in the winter, what service would you like to have (from this bike subscription)? | To prioritize which gadget options could Swapfiets offer to improve their service. Providing these gadget options could be one part of our marketing strategies to increase customer base especially during the challenging winter season. |
| **Sinus Milieus** | |
| What option best describes you? | To check with which milieu the participants identify and if there is any correlation between the milieu and the motivator from the previous question. |
| **Demographic** | |
| Howoldareyou? | To examine any tendencies in sinus milieus regarding the demographic difference. |
| What is your current occupation? | |
| Whatisyournationality? | |

**Raw Data**

| Participant | Interest | Motivator | Winter CyclingSuggestions | Sinus Milieus | Age | Occupation | Nationality |
|---|---|---|---|---|---|---|---|
| 1 | Yes | Minimalism (lesspossession) | Winter Tires, Face Mask, Earmuffs, Reflective Vest | Efficiency-oriented and "progressive and forward-thinking regarding consumerism" | 25 - 35 | Employed | German |
| 2 | Yes | Safety & Security (an always working bike) | Winter Tires, Gloves + Hand Warmers, Reflective Vest | Achieving and adapting to social order, "seeks security and harmony" | 36 - 65 | Employed | German |
| 3 | Yes | Less responsibility (of maintaining bike) | Winter Tires, Gloves + Hand Warmers, Rain Jacket | Efficiency-oriented and "progressive and forward-thinking regarding consumerism" | 36 - 65 | Employed | German |
| 4 | Yes | Minimalism (lesspossession) | Winter Tires, Face Mask, Earmuffs | Efficiency-oriented and "progressive and forward-thinking | 18 - 24 | Full-Time Student | Chinese |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | regarding consumerism" | | | |
| 5 | Yes | Safety & Security (an always working bike) | Gloves + Hand Warmers, Face Mask | Achieving and adapting to social order, "seeks security and harmony" | 36 - 65 | Employed | German |
| 6 | Yes | Minimalism (lesspossession) | Winter Tires, Face Mask, Reflective Vest, Insulated Water Bottle (metal) | Efficiency-oriented and "progressive and forward-thinking regarding consumerism" | 65+ | Retired | German |

**Calculation**

Also, with the use of both secondary data and survey results a potential market can be calculated of different age groups, nationalities and Milieus.

As a note, market share of Swapfiets cannot be calculated therefore, for the case of the calculation the share will be considered as 100%

Potential market population of 18 - 24 year olds

| Data | Value |
|---|---|
| Interest of 18 - 24 year olds in survey (%) | 63.83 |

| | |
|---|---|
| o.18 - u.25 year old population in Karlsruhe | 35,688 |
| o.18 - u.25 year old population in Karlsruhe excl. mountain villages | 34,622 |

Potential market population=Market share × Number of individuals interested in product/service

Potential market population=100%×(64.83%×35,688)

Potential market population≈22,446

≈23,137of 18 to 24 year olds in Karlsruhe

≈22,446of 18 to 24 year olds in Karlsruhe (excluding mountain villages)

Potential market population of Chinese 18 - 30 year olds

| Data | Value |
|---|---|
| Interest of Chinese in survey (%) | 90.91 |
| Chinese 18 - 30 year old population in Karlsruhe | 2,104 |

*For the sake of calculation, the ages of all the Chinese will be categorized from 18 - 35 to 18 - 30

<u>Example calculation of 18 to 30 year olds in Karlsruhe</u>

Potential market population=Market share × Number of individuals interested in product/service

Potential market population= 100% × (90.91% × 2,104)

Potential market population≈1,913

≈1,913of 18 to 30 year olds in Karlsruhe

(lack of secondary research does not allow for accurate calculations of the Chinese population in mountain villages)

Potential market population of Chinese students in KIT

(and the potential market population of Mandarin/Cantonese speaking nationality students in KIT)

| Data | Value |
|---|---|
| Interest of Chinese in survey (%) | 90.91 |
| Chinese students in KIT | 1,957 |
| Mandarin/Cantonese speaking nationality in KIT (China, Taiwan, Hong Kong, Singapore) | 1,999 |

Example calculation of Chinese students in KIT

Potential market population=Market share × Number of individuals interested in product/service

Potential market population=100%×(90.91%×1,957)

Potential market population≈1,780

≈1,780of Chinese students in KIT[47]

≈1,818of Mandarin/Cantonese speaking students in KIT (has a slight higher level of error due to the generalization of language and nationality)

[47]https://www.kit.edu/downloads/statistik_ws2019.pdf

Potential market population of different Milieus in Karlsruhe

| Data | Values of different Milieus | | | |
|---|---|---|---|---|
| | Escapist | New Middle-Class | High Achiever | Moversand Shakers |
| Interest (%) | 62.75 | 55.00 | 64.29 | 69.23 |
| Population in Karlsruhe[48] (%) | 15 | 8 | 27 | 10 |
| Population of Karlsruhe in general | 309,519 | | | |
| Population of Karlsruhe in general excl. mountain villages | 294,330 | | | |

Example calculation of market potential of escapists in Karlsruhe

Potential market population=Market share × Number of individuals interested in product/service

Potential market population=100% ×(62.75%×15%×309,519)

Potential market population≈29,314

[48]https://www.researchgate.net/figure/Abbildung-9-Sinus-Milieus-in-Karlsruhe-im-Jahr-2020_fig2_45828976

≈29,314of escapists in Karlsruhe

≈27,704of escapists in Karlsruhe excl. mountain villages

≈13,619of new middle-class in Karlsruhe

≈12,951of new middle-class in Karlsruhe excl. mountain villages

≈53,728of high achievers in Karlsruhe

≈51,901of high achievers in Karlsruhe excl. mountain villages

≈21,429of movers and shakers in Karlsruhe

≈20,377of movers and shakers in Karlsruhe excl. mountain villages

**Processed Data**

Market Potential Population

| Target Group | Potential Market Population in Karlsruhe[49] | Percentageof Karlsruhe Population (%) | Potential Market Population in Karlsruhe excl. mountain villages |
|---|---|---|---|
| 18 - 24 yearolds | 23,137 | 7.5 | 22,446 |
| 18 - 30 yearold Chinese | 1,913 | 0.6 | - |
| Chinese Students in KIT | 1,780 | 0.6 | - |
| Mandarin/Cantonese speaking students in KIT | 1,818 | 0.6 | - |
| Escapist | 29,314 | 9.5 | 27,704 |
| New Middle-Class | 13,619 | 4.4 | 12,951 |
| High Achiever | 53,728 | 17.4 | 51,901 |
| Moversand Shakers | 21,429 | 6.9 | 20,377 |

## The correlation between motivation and different Milieus

With the use of a stacked bar graph, a correlation between motivational factors and different milieus can be seen. Additionally, this proves the previous assumptions made and each motivation can be categorized into a different milieu. Furthermore, the number of each milieu and motivation interested in the service can be seen as only answers of individuals that were interested were questioned. While the graph supports both the assumptions and the hypothesis, the validity of the data is not very high due to the low sample size.

[49] https://web3.karlsruhe.de/Stadtentwicklung/statistik/pdf/2017/2017-bevoelkerung-jahresbericht.pdf

Different Milieus on winter cycling suggestions

| Gadgets | Escapist | New Middle-Class | High Achiever | Moversand Shakers |
|---|---|---|---|---|
| Winter Tires | 14 | 5 | **13** | 1 |
| Gloves + Hand Warmers | 12 | **6** | 5 | **8** |
| Face Mask | 11 | 4 | 8 | 3 |
| Earmuffs | **17** | 3 | 8 | 2 |
| Rain Jacket | 8 | 5 | 7 | 5 |
| ReflectiveVest | 15 | 5 | 8 | 1 |
| InsulatedWaterBottle (metal) | 7 | 3 | 8 | 3 |
| Other | 0 | 2 | 0 | 1 |

Through the comparison of different Milieus with their winter cycling suggestion, it can be seen that there is a difference in preference to the topic of winter cycling. Escapists seems to prefer earmuffs, the new middle-class and movers and shakers chose gloves and hand warmers, and high achievers picked winter tires. The previously used stacked bar graph can be used to give reasoning to the correlations and tendencies of Milieus on different winter cycling suggestions. An argument can be made for escapists that they prefer earmuffs so they increase their experience of cycling by decreasing the discomfort in the cold. On the contrary, the new middle-class seemed to have varying results but seemed to prefer security and safety with answers such as winter tires, gloves and a reflective vest. Next, high achievers also seemed to support the idea of minimalism with the majority of participants preferring winter tires rather than other direct possessions. Finally, movers and shakers seemed to not support the hypothesis and most preferred gloves and hand warmers rather than the expected winter tires. Overall, participants seemed to prefer winter tires, with a total of 33 answers.

**Analysis of Survey**

As seen from the processed data table above, there seem to be various trends in the population of Karlsruhe. First and foremost, results showed that all ages showed a similar number of interest in which each age group ranged in between 60 and 70 percent of interest. Furthermore, nationality had data which had a very low validity due to the low sample size of different nationalities. The only nationalities that had a sample size higher than 10 are the Germans and Chinese. The Germans had an interest of 59.74% (out of 77 participants) and Chinese had an interest of 90.91% (out of 11 participants). The Chinese population can be backed up and reasoned through the background research mentioned previously and the demographics of Karlsruhe from secondary research. In addition, all of the Chinese had the occupation of either a full-time student or an exchange student which also resulted in a very young aged demographic. Out of all the nationalities only this had such a specific and linear trend. When further talking about interest, different Milieus did not show a drastic difference in interest other than the new middle-class Milieu having a near 10 percent lower interest. On the other hand, the Milieus definitely did show a difference in motivation as mentioned previously which is important for coming up with marketing strategies.

Secondly, winter cycling was quickly touched upon by asking individuals what they wish to see out of a service to attract customers. Overall, there were no clear trends between the preference and other factors such as age, nationality and Milieu. While possible trends with the Milieu were mentioned, it seems unnecessary to draw a conclusion and a reasoning. It would be much more helpful by simply looking at the general preference, which is winter tires.

Finally, when calculating the potential market of different target groups in Karlsruhe, it can clearly be seen that the more general groups have a higher potential. While this makes sense, this also means that they have a higher uncertainty as there is a large difference in ratio between the demographics and survey results.

**Reflection**

When looking back at where the results come from, it is safe to say that the sample size does not represent the general population of Karlsruhe. More specifically, the nationalities did not make a large difference in the results, other than being able to see an interesting finding within the Chinese sample. The results can vary depending on different environmental factors of the research, such as the weather, location and time of the day. However, with this in mind, it can also be said that these factors did not make a great impact as we didn't look at data such as population density or frequency, rather than finding out the general demographic, the characteristics and opinions of individuals were the main aim of this study.

While participants and the research environment might have played a big role in creating errors within the results, researchers could also be seen as another factor. As a matter of fact, many things were not kept constant within this survey. The researcher differed depending on the time and location, this could have played a large factor. Also, some participants were surveyed in German, others in English and finally, some participants might have been pushed into answering certain answers, or given the feeling to do so.

In conclusion to this survey, while many errors existed, a lot of valuable data was collected. This data not only supports the aim of the research but is very valuable to be used in understanding the cycling market of Karlsruhe.

## Focus on Escapist milieu

For our marketing strategy, we solely focus on the Escapist milieu.The reasons are:

- Greatest market potential of this milieu
- Time frame of the project
- Given marketing budget by Swapfiets.

## MARKETING MEDIUM DECISION MAKING

With the decision our group made with the main target group of escapist, their main motivators were taken into consideration when creating suitable marketing strategy: fun and experience, and less responsibility regarding maintenance of the bike.

**4Ps of marketing mix**

In order narrow down the best possible choice for marketing promotion to achieve the goals, we have used the method of marketing mix of 4Ps and scrutinize the product.

**Product**:

Karlsruhe Swapfiets offers Deluxe 7, which is their main product, the city bike. Deluxe 7 has seven gears and has double locks provided. The significant blue front tire represents the brand, Swapfiets.[50]The company offers a monthly based subscription, which includes all the services with bike repair in case the bike malfunctions.

**Price**:

The customers are supposed to subscribe to their monthly subscription. For Deluxe 7, which is the only product of Karlsruhe Swapfiets, it is €19,50  per month. However, the company also offers a discount for students by €17,50 per month.

**Place**:

In Karlsruhe, Swapfiets has its store located in the city center where they exhibit their bikes and customers can pick up their bikes and visit to ask for service.Also, in order to register for subscriptions, online Swapfiets website is also available. These two channel options are where customers can access to Swapfiets' product.

**Promotion**:

For further promotion of the product, our group has taken some steps to come up to the possible marketing strategies and campaigns that best suits our main target group, the escapists.

[50]https://swapfiets.de/en/bike/karlsruhe/de-dlu7-norm-upgraded

First of all, the overall approach should be executed locally, especially Karlsruhe, as the company wants to focus mainly on this city. Accordingly, the social media channel of Swapfiets, such as Instagram, Facebook, or Twitter, would not be the main channel for this marketing campaign since they are not only for local but are for whole Germany. Furthermore, radio and TV are not considered because of the high costs involved, which does not fit into the given marketing budget.

By considering these options, we concluded that two possible best marketing campaign for Swapfiets would be:

- **1. Poster advertisements**
- **2. Guerilla marketing (Mural painting or street art on the walls + Swapfiets bike parade)**

## 1. <u>Poster advertisements</u>

In order to further provoke the demand of bikes within our target group customers, we reached out to the idea of posters. This poster campaign focuses on another characteristics of the escapists- willingness to worry less about the maintenance of the bike when it's broken or loss. Taking into this consideration, we found that there is a matching components of Swapfiets service, "no hassle with repairs". As a result, we proposed to target the escapist with the message of "No worries, we are Swapfiets" on the poster. Our concrete idea of such campaign is to create one poster where people from different countries can be seen, saying the sentence above, in their native language. This also emphasizes the Swapfiets community being quite international and diverse.

As for that, it does not only address our main target group of escapists, but also affects the exchanges students in Karlsruhe as a side effect. These exchange students are the sub-target group from the Movers and Shakers. This consequence could be derived from the international aspect of the poster with multiple languages and nationalities. Also, Chinese population is quite large in Karlsruhe, and by having Chinese in the poster, we are including them from the High Achiever milieu.

In order to surely implement the poster advertisement around the city, there are possible locations where the company could consider to hang their posters. For example, at the main and central stations where reaching young, mobile, and active target group could be easily reached. Also, around the city, general advertising column could be used reliably especially for every cultural and brand message.

Cost Proposal[51]:

- City-Light-Poster
  - € 10,70 per day
  - One of the most used poster advertisement channel in Karlsruhe

- Allgemeinstelle
  - From €0,85 to €1,38per day
  - One of the most seen advertisement channel in Karlsruhe

[51] https://www.stroeer.de/nc/aussenwerbung/werbemedien/kategorie/plakatmedien/werbemedien/Product/list.html

- Großfläche Bahnhof
    - €8,25  per day
    - Countless number of people are always there at the main central stations

## 2. Guerrilla marketing

For Swapfiets to focus on low-cost and unconventional marketing strategy with an element of surprise, *guerrilla marketing* would be an efficient marketing campaign. The surprising tactics to approach the people of Karlsruhe would create a memorable impression.
With targeting the escapist milieu, creativity and boldness would enable them to admire Swapfiets promotion. One type of guerrilla marketing to address the escapist audience would be 'street marketing'.

### Mural painting

Public street marketing could be presented by Swapfiets in two forms: an interactive mural and/or 3D street chalk art. With an interactive mural, a 'free wall' would need to be searched and provided in the city of Karlsruhe, and a Swapfiets bike itself would be the interactive art. An inspirational example would be the "Kids on bicycle" mural by Ernest Zacharevic in Malaysia.

"Kids on bicycle" Ernest Zacharevic (2012)

### 3D street art

3D street chalk art is another great option for temporary public art that may allow themed art for the holidays during winter seasons. The 3D aspect allows a fun and interactive way for people to take photos with the Swapfiets campaign.
Costa Coffee captured attention with 3D street art by Manfred Stader. The unique chalk drawing grabs fast attention from people walking by.

In the city of Karlsruhe, to create the interactive murals and/or 3D street art marketing would be with the establishing alliances, such as the ADFC for promoting cycling. To apply these street art approaches, they may be implemented any time of the year. To possibly promote winter cycling, a holiday themed 3D temporary street art may attract people during the months of November and December. Those being involved in this artistic campaign, would be to launch a creative team in which establishes alliances with local and/or regional artists. The locations of the street art campaigns would be placed in the highest foot traffic areas of Karlsruhe, such as the Karlsruhe Palace and garden, and Marktplatz area. This campaign targets the 'escapist milieu' due to the fun and lively approach of marketing. As for estimated costs, to invest in the creative team, design, materials, etc… would be depended on those variables. The larger scope of this allowed project would be a larger wall for the mural.

Costa Coffee 3D art by Manfred Stader (2013)

## Cycling Parade

A cycling parade would enable Swapfiets to approach individuals that are outside in public, but would also create a fun and exciting environment, ideal for people part of the escapist milieu. The difficult part of a parade is the planning and setting up. The main procedure of this parade will be done with participants on Swapfiets bicycles, riding together through the city. Ideally this would be at Marktplatz or near the palace due to the dense population and amount of people available. The most convenient dates to do this parade on, is June 3rd, which is world bicycle day, and shortly before Christmas. Doing the parade before Christmas season could help with the increase of cycling in the winter. An incentive that could attract people to participate might be a buffet or some kind of food event at the store. While the incentive can vary greatly, as long as participants are willing to take part in the parade, it should be

enough. Costs will differ greatly depending on the incentive. Additionally, bicycles will need to be provided by Swapfiets depending on the number of participants. Overall the end costs of this parade will depend on the number of participants. With a probable participant number of 20, food and drink costs could add up to approximately €250.

## Winter cycling

Nowadays, more and more people use the bicycle for the way to work, shopping or leisure activities all year around. However, cycling in winter is only safe when the right equipment is used for both the cyclist and the bicycle. The ADAC, for instance, recommends cyclists to make use of:

- Good, not worn tread of the tires (best option would be special winter tires)
- Clear illumination
- Weatherproof & light clothing
- Gloves
- Etc.[52]

Consequently, it is necessary to consider the need of this special winter equipment in order for Swapfiets to maintain their current customers and preferably increase the customer base by attracting new potential customers.Therefore, many solutions to different problems were researched which include:

| **Problem** | **Solution** |
|---|---|
| Slippery roads (rain and snow) | Winter tires, metal-studded tires |
| Coldtemperature | Gloves, hand warmers, face mask, ear warmers, insulated water bottle, heated seats, hot pack |
| Rain | Rain jacket |
| Unnoticeable/moredangerous (dark) | Reflective vest, more lights/reflectors |

[52]https://www.adac.de/verkehr/verkehrssicherheit/wetter/fahrrad-winter/

These are various ideas that Swapfiets could implement to attract customers. According to the survey conducted, the majority seem to prefer winter tires but there was no real difference, whether individuals prefer their comfort or their safety in the winter. However, it seemed as though individuals prefer gadgets and additions that they do not own, meaning options such as better tires and reflective vests will always help, since it is not commonly owned. Coming back to ADAC, they also suggest having a good maintenance of the bike with tires that work on slippery roads.

In conclusion, the cycling numbers in winter decrease due to riskier environments and discomfort. While Swapfiets also experiences the loss in customers during the season, many possible solutions are provided to combat those problems. Some of which include winter tires, reflective vests and additional clothing for warmth.

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